Submission Readiness Report

Aligning cybersecurity investment with business-impact risk: A knowledge-based AI decision-support model evaluated on real-world incident data

Author: Rahul Chandra Nandagiri | Target journal: Knowledge-Based Systems (Elsevier; SJR 2025 Q1) | Article type: Short Communication | Report date: 23 October 2026 (updated 24 October 2026)

This report gives the seven outputs required by the DBA 802 "Final Prompt" (journal template compliance and final manuscript editing). Every check marked Pass was actually performed. Items that need the author are marked Author or Check and listed in Section 6.

1. Revised journal-formatted manuscript and submission package

- **Manuscript.docx** and **Manuscript.pdf** – the paper in a regular reading layout (1.15 line spacing, no line numbers, 11 pages). The journal received the same text in its review layout (double spacing, continuous line numbers).
- **Highlights.docx**, **Graphical_abstract.png/.pdf**, **Figure_1.png/.pdf**, **Figure_2.png/.pdf**, **Table_1.csv**, **Table_2.csv**.
- **Supplementary_Material.docx** (Tables S1–S10) and **Supporting_Data_and_Code.zip** (code, all result tables, figures, analysis cohort, pinned VCDB archive with license notice).
- **Cover_Letter.docx** for the editor.
- **GitHub repository** (fixed and pushed): https://github.com/rahulchandranandagiri/cybersecurity-investment-business-performance

2. Journal template compliance checklist

Source of requirements: Knowledge-Based Systems Guide for Authors (PDF posted on the course dashboard, 23 October 2026) and the accepted reference paper (KNOSYS_116815), used for presentation style only.

Requirement	Status	Evidence / note
Article type: Short Communication (≤10 double-spaced pages incl. tables and figures)	Check	In the double-spaced file sent to the journal, the main text with both tables and figures runs from page 3 to page 12, about 10 pages. That file is 17 pages because it also holds the title page, abstract, declarations and 36 references. The guide does not say whether those count, so this item is flagged in Section 6.
Editable source file (.docx), single-column layout, no strikethrough/underline	Pass	Manuscript.docx (Word), single column, US Letter, 12 pt Times New Roman, page numbers. The file sent to the journal uses the review layout (double spacing, continuous line numbers). The copy posted on the course page and GitHub uses a regular reading layout (1.15 line spacing, no line numbers, 11 pages) with identical text, tables, figures and references.
Title page: concise title, author names, affiliation with superscript letter, corresponding author with e-mail	Pass*	Present. *The postal address of the affiliation must be confirmed by the author (Section 6).
Abstract ≤250 words, stand-alone, no references, abbreviations defined	Pass	245 words; KB-CIDS, BIRR and VERIS are defined at first use.
Keywords: 1–7, English, no "and/of" compounds	Pass	5 keywords (the Elsevier submission form accepts at most 5).
Highlights: 3–5 bullets, ≤85 characters each, separate editable file with "highlights" in the name	Pass	Highlights.docx; 5 bullets of 75–79 characters.

Requirement	Status	Evidence / note
Math as editable text, displayed equations numbered, variables in italics, solidus for small fractions	Pass	Seven Office Math equations, Eqs. (1)–(7); exponent 1/(β+1) uses a solidus.
Tables: editable, captioned, cited, numbered, notes below, no vertical rules or shading	Pass	Tables 1–2 are Word tables with horizontal rules only; Tables S1–S10 in the supplementary file.
Figures: separate files with logical names, cited in order, captions, adequate resolution, accessible color	Pass	Figure_1 and Figure_2 as 600 dpi PNG plus vector PDF; color-blind-safe palette; also embedded in the manuscript for review.
Generative AI and figures policy	Pass	Figures are data visualizations generated by the analysis code from the data; captions state this. No generative image tools used.
Section numbering (1, 1.1, ...); abstract not numbered	Pass	Sections 1–6 with numbered subsections.
References: numbered in citation order, [n] in text, all real, DOIs, journal abbreviations, every item cited	Pass	36 references, all cited in order (automated check); dataset tagged [dataset]; see Section 5.
Research data (Option C): deposit, cite and link data; data statement	Pass	Public VCDB data pinned to a commit and cited; code and derived data on GitHub; data availability section included.
Declaration of competing interest	Pass*	Statement included. *The Elsevier declarations tool must also be completed online at submission (author action).
Funding statement	Pass	Standard "no specific grant" sentence.
CRediT author contribution statement	Pass	Included.
Declaration of generative AI use (separate section before references)	Pass	Included with the required wording.
Graphical abstract	Pass	Not mandated by the KBS guide but required by the course; Graphical_abstract.png (2385×984 px, 300 dpi) and PDF provided.
Supplementary material cited with captions	Pass	Supplementary_Material.docx (Tables S1–S10) cited in Sections 2.2, 4.1 and 4.2.
Language: one variety of English, inclusive language	Pass	American English throughout; neutral language.
Submission declaration (original, not under consideration elsewhere)	Author	Stated in the cover letter; the author must confirm when submitting.

3. Summary of scientific and editorial improvements

- **From an empty repository to a working study.** Every module in the GitHub repository was 0 bytes. It now holds a complete, deterministic pipeline (data download with checksum, cohort rules, knowledge base, six models, decision layer, evaluation, figures). One command, run_experiment.py, regenerates every number, and the work went up in five commits.
- **From a conceptual framework to a tested model.** Research Paper 1 argued conceptually that cybersecurity value depends on aligning investment with business exposure. KB-CIDS turns that idea into a computable knowledge-based decision model and tests it out of time on 9,284 real incidents.
- **Research gap and novelty.** The introduction now separates analytical investment models from data-driven cyber-risk prediction and shows that the two streams are rarely linked or validated out of time. The three contributions are stated without overclaiming.
- **Methodology and equations.** The business-impact outcome, Gordon–Loeb response, budget program, closed-form allocation, weighting scheme and evaluation metric are given as Eqs. (1)–(7) and match the code. Algorithm 1 summarizes the procedure.

- **Experimental rigour.** Development to 2016, evaluation from 2017. Hyperparameters were tuned on an inner validation split only. There are 6 predictive and 7 decision baselines plus an oracle, bootstrap confidence intervals, paired tests, an ablation, 6 robustness specifications and 9 parameter settings.
- **Honest interpretation.** A negative result (AI imputation did not transfer out of time) is reported and explained rather than hidden. Limitations cover data selection, the proxy outcome and the expert-defined rules.
- **Business implications.** Section 4.3 turns the findings into three managerial actions and an auditable rationale example (small healthcare organizations).
- **Language and format.** The text is concise, uses American English and consistent terms (MBI, BIRR, KB-CIDS) and follows the KBS reference-style example. The structure mirrors the accepted reference paper (introduction, model, design, results, limitations, conclusion, declarations).
- **References.** 36 references, every one checked against Crossref/OpenAlex/publisher records. 31 come from Q1 journals published by Elsevier (ScienceDirect), IEEE and MDPI.

4. Verification of the two figures and two tables

Item	Generated by	Checks performed	Result
Fig. 1	explainability.figure_1 ← S9 allocations	600 dpi PNG (5179×3155 px) + PDF; 3 panels share axes; values ≥15% labeled; color bar and family legend; caption matches content	Pass
Fig. 2	explainability.figure_2 ← S4 budget curve, S2 profile scores	600 dpi PNG (4531×2074 px) + PDF; 95% CI band; equal-allocation reference; legends do not overlap data	Pass
Table 1	Table_1_predictive_validity.csv	All cells exported directly from the results file (no manual transcription); 2,000 bootstrap resamples; paired AUC tests	Pass
Table 2	Table_2_decision_performance.csv	All cells exported directly; 1,000-draw CIs; Wilcoxon tests across 18 profiles	Pass
Numbers in text	paper/verify_manuscript.py	81/81 automated checks passed (limits, citations, every quoted number against the results files)	Pass
Reproducibility	run_experiment.py on two library stacks	Identical point estimates (all Table 1 metrics, all BIRR values, Wilcoxon p-values, robustness and sensitivity tables) with numpy 2.5/pandas 3.0/scikit-learn 1.9 and with numpy 2.1/pandas 2.2/scikit-learn 1.5; bootstrap intervals differ only by Monte Carlo error	Pass

5. Reference and citation verification report

All 36 references were checked against bibliographic records (Crossref, OpenAlex, RePEc or the publisher/institutional record). Titles, authors, volumes, pages or article numbers and DOIs match. All 36 are cited in the text in numerical order of first appearance, and no citation points outside the list (automated check). **31 of 36 references come from Q1 journals of the three requested publishers: 21 Elsevier/ScienceDirect, 6 IEEE and 4 MDPI.** Quartiles were checked on SCImago (SJR 2025, best quartile). The other five are kept because scientific integrity requires them: the foundational Gordon–Loeb model (ACM), the two standards the knowledge base is built on (CIS Controls v8, NIST CSF 2.0), the dataset (VCDB) and the software (scikit-learn).

#	First author (year)	Publisher	Journal / type	SJR quartile	DOI / URL	Verified via
[1]	Vial (2019)	Elsevier	J. Strateg. Inf. Syst.	Q1 (2025)	10.1016/j.jsis.2019.01.003	Crossref
[2]	Verhoef et al. (2021)	Elsevier	J. Bus. Res.	Q1 (2025)	10.1016/j.jbusres.2019.09.022	OpenAlex
[3]	Kamiya et al. (2021)	Elsevier	J. Financ. Econ.	Q1 (2025)	10.1016/j.jfineco.2019.05.019	Institutional repository

#	First author (year)	Publisher	Journal / type	SJR quartile	DOI / URL	Verified via
[4]	Bianchi & Tosun (2021)	Elsevier	Int. Rev. Financ. Anal.	Q1 (2025)	10.1016/j.irfa.2021.101795	OpenAlex
[5]	Rodrigues et al. (2024)	MDPI	Future Internet	Q1 (2025)	10.3390/fi16060201	OpenAlex
[6]	Eling & Wirfs (2019)	Elsevier	Eur. J. Oper. Res.	Q1 (2025)	10.1016/j.ejor.2018.07.021	RePEc
[7]	Podrecca et al. (2022)	Elsevier	Comput. Ind.	Q1 (2025)	10.1016/j.compind.2022.103744	Crossref
[8]	Hasan et al. (2021)	Elsevier	J. Inf. Secur. Appl.	Q1 (2025)	10.1016/j.jisa.2020.102726	OpenAlex
[9]	Demek & Kaplan (2023)	Elsevier	Int. J. Account. Inf. Syst.	Q1 (2025)	10.1016/j.accinf.2023.100616	OpenAlex
[10]	Gordon & Loeb (2002)	ACM	ACM Trans. Inf. Syst. Secur.	n/a (foundational; not a target publisher)	10.1145/581271.581274	OpenAlex
[11]	Huang & Behara (2013)	Elsevier	Int. J. Prod. Econ.	Q1 (2024)†	10.1016/j.ijpe.2012.06.022	RePEc
[12]	Nagurney & Shukla (2017)	Elsevier	Eur. J. Oper. Res.	Q1 (2025)	10.1016/j.ejor.2016.12.034	OpenAlex
[13]	Simon & Omar (2020)	Elsevier	Eur. J. Oper. Res.	Q1 (2025)	10.1016/j.ejor.2019.09.017	OpenAlex
[14]	Paul & Wang (2019)	Elsevier	Decis. Support Syst.	Q1 (2025)	10.1016/j.dss.2019.05.009	OpenAlex
[15]	Chronopoulos et al. (2018)	IEEE	IEEE Access	Q1 (2025)	10.1109/ACCESS.2017.2773366	OpenAlex
[16]	Brho et al. (2025)	Elsevier	Int. J. Prod. Econ.	Q1 (2024)†	10.1016/j.ijpe.2024.109448	Crossref
[17]	Fielder et al. (2016)	Elsevier	Decis. Support Syst.	Q1 (2025)	10.1016/j.dss.2016.02.012	Institutional repository
[18]	Fernandez De Arroyabe et al. (2023)	Elsevier	Comput. Secur.	Q1 (2025)	10.1016/j.cose.2022.102954	Crossref
[19]	Owoleso et al. (2026)	MDPI	J. Cybersecur. Priv.	Q1 (2025)‡	10.3390/jcp6050171	OpenAlex
[20]	Xu et al. (2018)	IEEE	IEEE Trans. Inf. Forensics Secur.	Q1 (2025)	10.1109/TIFS.2018.2834227	OpenAlex
[21]	Hammouchi et al. (2023)	IEEE	IEEE Trans. Dependable Secure Comput.	Q1 (2025)	10.1109/TDSC.2022.3149208	OpenAlex
[22]	Seh et al. (2020)	MDPI	Healthcare	Q1 (2025)‡	10.3390/healthcare8020133	OpenAlex
[23]	Yankson et al. (2025)	MDPI	J. Cybersecur. Priv.	Q1 (2025)‡	10.3390/jcp5030070	OpenAlex
[24]	Buczak & Guven (2016)	IEEE	IEEE Commun. Surv. Tutor.	Q1 (2025)	10.1109/COMST.2015.2494502	Crossref
[25]	Xin et al. (2018)	IEEE	IEEE Access	Q1 (2025)	10.1109/ACCESS.2018.2836950	OpenAlex
[26]	Barredo Arrieta et al. (2020)	Elsevier	Inf. Fusion	Q1 (2025)	10.1016/j.inffus.2019.12.012	Crossref
[27]	Capuano et al. (2022)	IEEE	IEEE Access	Q1 (2025)	10.1109/ACCESS.2022.3204171	OpenAlex
[28]	Jia et al. (2018)	Elsevier	Engineering	Q1 (2025)	10.1016/j.eng.2018.01.004	OpenAlex
[29]	Sarhan & Spruit (2021)	Elsevier	Knowl.-Based Syst.	Q1 (2025)	10.1016/j.knosys.2021.107524	OpenAlex
[30]	Zhao et al. (2022)	Elsevier	Knowl.-Based Syst.	Q1 (2025)	10.1016/j.knosys.2021.108086	OpenAlex
[31]	Moreno-Torres et al. (2012)	Elsevier	Pattern Recognit.	Q1 (2025)	10.1016/j.patcog.2011.06.019	OpenAlex
[32]	Beaman et al. (2021)	Elsevier	Comput. Secur.	Q1 (2025)	10.1016/j.cose.2021.102490	OpenAlex
[33]	Center for Internet Security (2021)	CIS	Standard	n/a (standard used by the knowledge base)	https://www.cisecurity.org/controls/v8	Publisher web page
[34]	NIST (2024)	NIST	Standard	n/a (standard used by the knowledge base)	10.6028/NIST.CSWP.29	DOI record
[35]	Verizon RISK Team (2026)	Verizon	Dataset	n/a (data source)	https://github.com/vz-risk/VCDB	GitHub commit
[36]	Pedregosa et al. (2011)	JMLR	Software paper	n/a (software citation)		JMLR BibTeX record

† SCImago's 2025 page for IJPE could not be retrieved during this session; Q1 in all four categories in SJR 2024 (SCImago ID 19165). ‡ Best quartile: J. Cybersecur. Priv. is Q1 in Computer Science (misc.) and Q2 in Artificial Intelligence; Healthcare is Q1 in Leadership and Management and Q2 in its health categories. MDPI's Mathematics and Systems journals were checked and excluded because they are Q2.

6. Unresolved scientific or formatting issues (author actions; status on 24 October 2026)

- **Page limit.** The Guide caps Short Communications at about 10 double-spaced pages including tables and figures. The main text sits at that limit, but the full double-spaced file (with references) is 17 pages. Option A: submit as a Short Communication and, if the editor asks, move Algorithm 1 and the robustness paragraph to the supplementary file. Option B: submit as an Original Research paper (limit about 20 pages). **Decision:** submitted as a Short Communication (Option A).
- **Author list and affiliation.** Confirm the author list (for example, whether the course instructor should be a co-author) and the full postal address of Midwest University before submitting. Elsevier does not allow authorship changes after acceptance and only allows them before acceptance through a formal request. **Confirmed at submission:** single author; affiliation Midwest University, Wentzville, Missouri, United States.
- **Declarations tool.** Complete Elsevier's online declarations tool during submission and upload the Word file it generates. **Done at submission:** no competing interests confirmed in the submission form.
- **Journal submission and screenshot. Done:** submitted to Knowledge-Based Systems on 24 October 2026 (status "Submission Received"); the screenshot is posted on the course page.
- **AI-imputation component.** Reviewers may ask why a component that did not help out of time is kept. The manuscript explains this as a reporting-bias finding. If reviewers prefer, the full model can be re-specified without imputation in a revision and re-validated.
- **Knowledge-base validity.** The 22 rules are expert-defined (drafted with AI assistance and grounded in CIS v8/NIST CSF 2.0). Practitioner validation is recommended and is listed as a limitation.
- **Outcome proxy.** BIRR is a model-based proxy, not observed financial loss. This is stated in the limitations; reviewers may request loss-magnitude data in a later study.
- **Data deposit DOI (optional).** For full Option C compliance, consider archiving a GitHub release on Zenodo to obtain a DOI and cite it in the data availability statement.
- **Author review.** Elsevier holds authors accountable for AI-assisted content. Read the manuscript fully, check every claim, and update the generative AI declaration if other tools (for example ChatGPT or Cursor) are used later.
- **Licenses.** Code is released under MIT and derived data under CC BY-SA 4.0 (required by the VCDB license). Change the code license if you prefer another one.

7. Final recommendation

Minor Revisions (author actions only). The science, analysis, figures, tables, references and journal formatting are ready. All automated checks pass, and the code reproduces every number on two library stacks. Before submission the author must (i) choose the article type given the page limit, (ii) confirm the author list and affiliation address, (iii) read and approve the full text, and (iv) complete the online declarations. After that the manuscript is **Ready for Submission**.

Verification was repeated after the last substantive edit: 81/81 checks passed, and the full pipeline was re-run three times with identical results.

Update, 24 October 2026. Author actions completed; submitted to Knowledge-Based Systems as a Short Communication. The course and GitHub copy is set in a regular reading layout (no line numbers, 1.15 spacing, 11 pages); its text is identical to the submitted file and 81/81 checks still pass.